LibreLift - Design Document

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Reviewers	[who reviewed it — a teammate, the instructor, a design-review partner]
Status	In review
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Links	proposal · repository · issue tracker

Revision history

Date	Author	What changed
10-16-26	James Teuscher	First draft
10-20-26	James Teuscher	First full draft
10-21-26	James Teuscher	Finished first copy
11-4-26	James Teuscher	Revise to respond to feedback

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1. Context and scope

LibreLift is a native mobile app for planning, logging, and tracking workouts for those who want to use their phones to do so. It aims to provide the basic functions of creating reusable workout templates, using a template for a workout and logging reps and weight, and reviewing the details of past workouts. For more information about the context of this project, see the proposal document.

2. Goals and non-goals

Goals

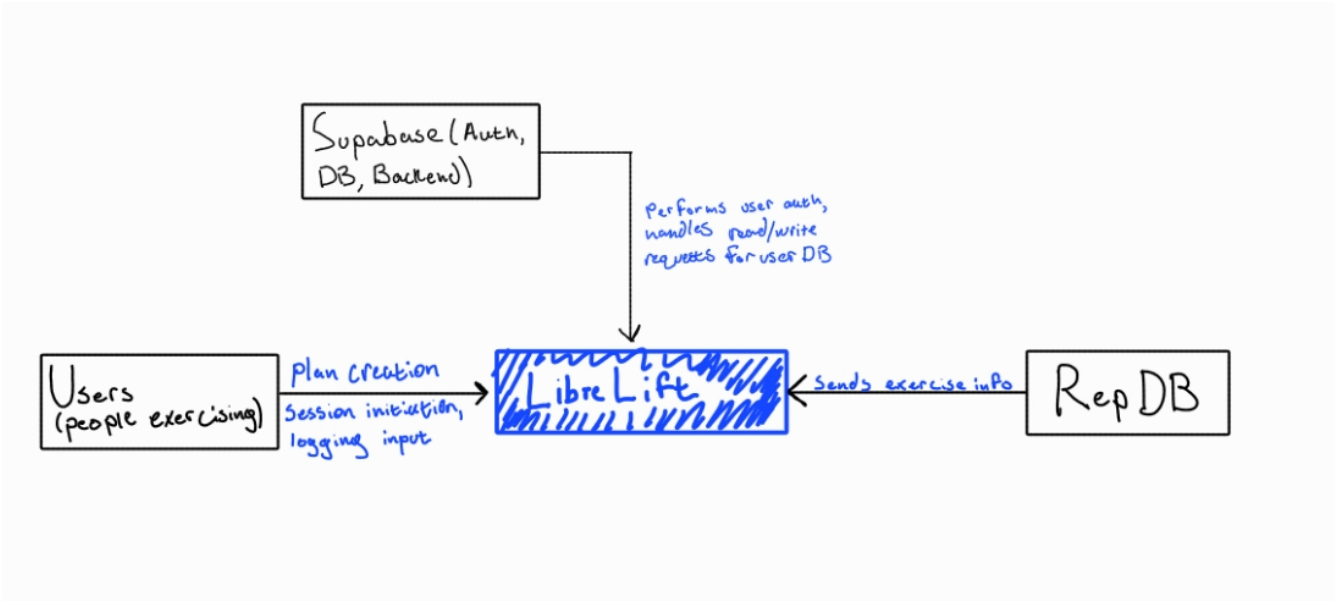
- Connect to RepDB, an external exercise DB, and load the catalog of ~500 free exercises into the app for use. Besides name, the exercises in the app should be identifiable by target muscles and equipment.
- Support creation, persistence, editing, and deletion of reusable exercise templates.
- Support starting, stopping, and completing workout sessions for an existing template. Workout session allow logging repetitions performed and weight used for each exercise in a plan.
- Support storing "one-rep" maximum weight for certain exercises, and enter weight in template exercises as a percentage of that maximum.

Non-goals

- Social exercising features such as followers/friends, sharing workouts, leaving comments, etc., are excluded. They fall outside the scope of primary user needs.
- No body composition and fitness goal customization.

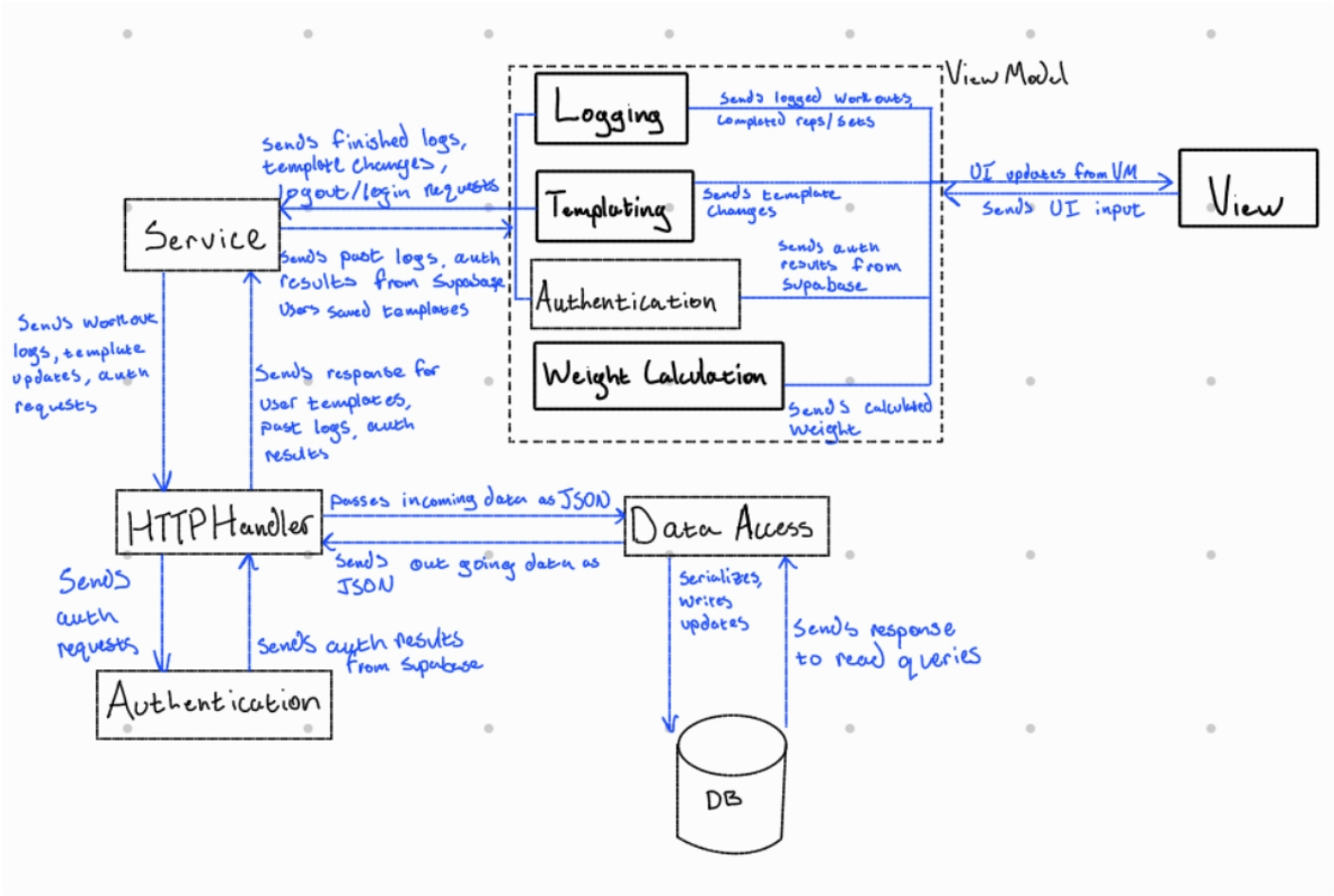
3. System-context diagram

There will only be one type of user interacting with this system. They will create plans and log workouts. The other external component interacting with this system is the CDN/API that provides a catalog of exercises. It will interact with the system by providing the exercise catalog via HTTP.



4. Architecture and components

Component	Responsibility	Talks to	Interface (rough)
View	Displays UI and accepts user input	ViewModel	function calls, touch screen elements
ViewModel	Business logic of logging, templating, and weight calculation	View, Service	function calls
Service	Makes calls to backend to read/write data for ViewModel operations	ViewModel, HTTPHandlers	function calls, HTTP
HTTPHandlers	Accepts incoming HTTP calls from frontend, handles request payloads	Service, DataAccess, Authentication	function calls, HTTP
Authentication	Accepts authentication requests via HTTP, uses Supabase auth to accept or reject	HTTPHandler	function calls, HTTP
DataAccess	Adjusts HTTP payloads to be ready for DB reads/writes	DB, HTTPHandlers	function calls, HTTP
DB	Stores data persistently, supplies data for app UI	DataAccess	HTTP



Core flow - Making a workout template

1. A user logs in or signs up with credentials
2. User taps the screen to create a plan
3. View layer sends input to ViewModel
4. ViewModel navigates to template creator business logic, calls service to load catalog
5. Service calls the Exercise CDN to load catalog, returns data to ViewModel
6. ViewModel sends exercises to View
7. View displays exercises
8. User chooses exercises, weight, reps
9. View sends selections to ViewModel
10. ViewModel saves plan in-memory as user builds template
11. User taps "Save plan"
12. View call ViewModel's save function
13. ViewModel sends plan data to service via function call
14. Service serializes the plan into JSON, calls backend via HTTP
15. HTTPHandler receives plan JSON, passes to DataAccess layer via function call
16. DataAccess layer deserializes JSON, writes to DB via HTTP

The failure case

1. User taps the screen to create a plan
2. View layer sends input to ViewModel
3. ViewModel navigates to template creator business logic, calls service to load catalog
4. Service checks if the catalog is cached. If not, call the Service layer
5. Service calls the Exercise CDN to load catalog, returns data to ViewModel
6. CDN throws an error and does not return catalog
7. Service passes an error to the ViewModel layer
8. ViewModel layer formats the error message to display on the UI
9. UI displays the error message for the User, asks to wait for a period and then try again

Core flow - Logging a workout:

1. A user logs in using credentials
2. User selects one of their templates and taps a "Start workout" button.
3. A stopwatch/counter displays workout duration, and lists the exercises in the workout.
4. For each exercise, the user is able to tap it and see a detailed view
5. The user can adjust actual weight and reps used if it differs from the templated amounts
6. After doing a one round of exercises, the user clicks complete, and the next round is presented.
7. This is repeated for all the templated amount of rounds, for each exercise.
8. When finished, the user taps "Finish workout"
9. The workout is completed and added to the User's log
10. The user can navigate to a history/log screen and see the details of their session

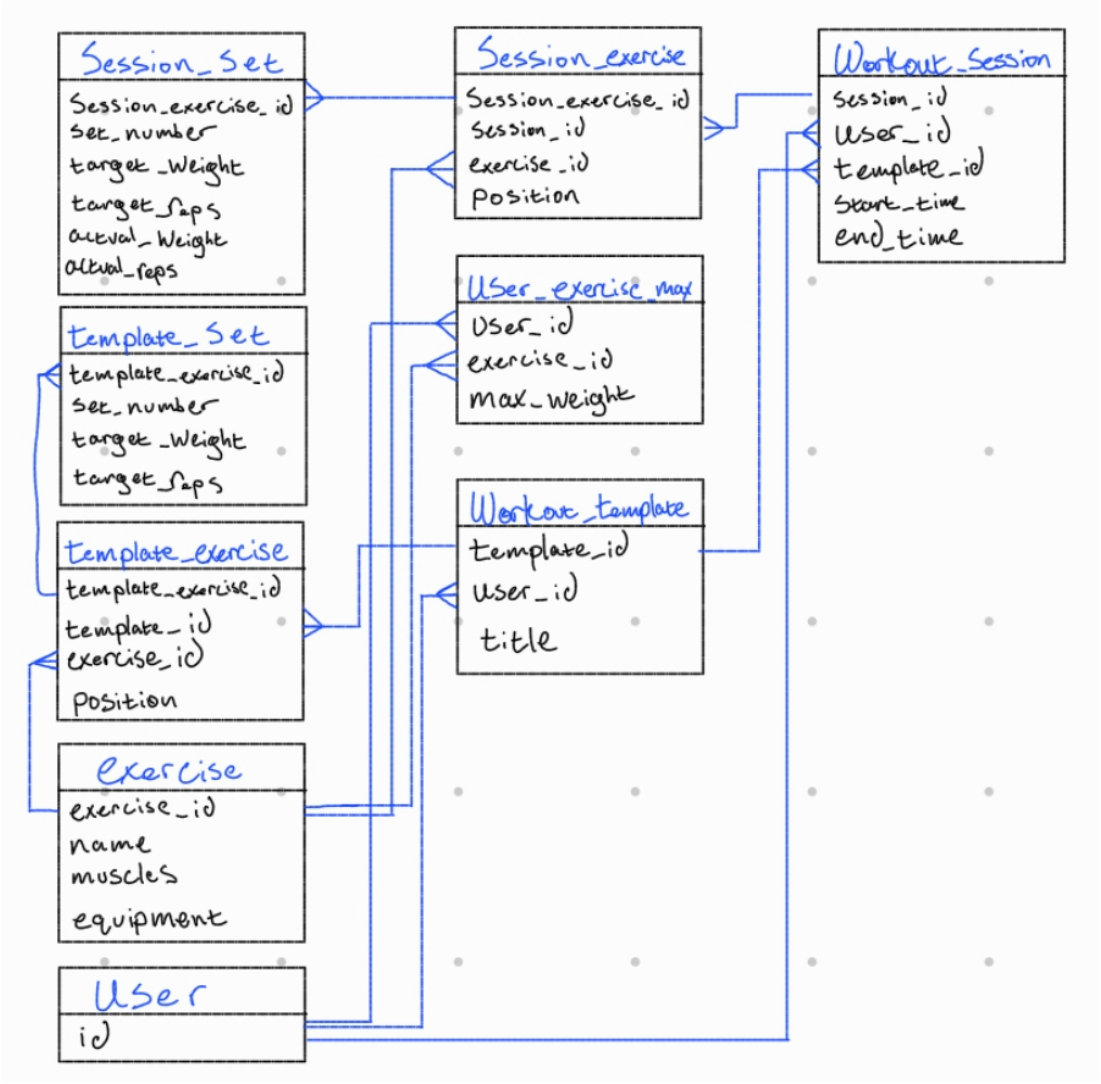
The failure case

1. A user logs in using credentials
2. User selects one of their templates and taps a "Start workout" button.
3. The user proceeds through the workout as stated above
4. When tapping the "Finish workout" button, the DB write fails
5. The error message is sent back to the user and told to try again in a few minutes.
6. If the error persists a second time, the user is presented a button to send feedback.
7. The feedback link sends an email to the maintainers with the user input.

5. Data model

Entity	Key fields	Relationships	MVP or roadmap
User	id	has workout templates, has workout sessions, and has exercise max weights	MVP
Exercise	exercise_id, name, muscles, equipmenet	has a User max weight, part of template exercise and session exercise	MVP
Workout template	userId, template_id, title	belongs to User, has exercises, fills a workout session	MVP
Template exercise	template_exercise_id, exercise_id, template_id, position	part of Workout template, is an exercise, has sets	MVP
Template set	template_exercise_id, set_number, target_weight, target_reps	part of Template exercise	MVP
User exercise max	user_id, exercise_id, max_weight	belongs to User, describes exercise	MVP
Workout session	session_id, user_id, template_id, started_at, completed_at	belongs to user, filled by workout template, contains session exercises	MVP
Session exercise	session_exercise_id, session_id, exercise_id, position	part of Workout session, has session sets	MVP
Session sets	sesson_exercise_id, set_number, target_weight, target_reps, actual_weight, actual_reps	part of session workout	MVP

ERD:



6. Technology choices and alternatives considered

Decision: What ExerciseDB/CDN will be used

- **Chosen:** RepDB
- **Alternative considered:** yuhonas/free-exercise-db
- **Why the chosen option wins, given our goals:** There is a free starter license, and the example images are superior than those in free-exercise-db
- **Evidence:** The spike utilized RepDB, while a personal side project has utilized free-exercise-db, and the RepDB images are superior, while muscle grouping, exercises available, and instructions are roughly equivalent.
- **What would change our mind:** If the user base grows enough that the RepDB free license is no longer usable, I would want to switch to the free alternative of free-exercise-db

Decision: Which Backend as a Service (BaaS) technology will be used?

- **Chosen**: Supabase
- **Alternative considered**: Firebase
- **Why the chosen option wins, given our goals**: 2 main reasons: First, Supabase is better build for open-source contributions, since contributors can run a local version of the backend without needing access to the production credentials. Second, the data model for LibreLift (workouts, exercises, template vs session) fits well in a relational model, which is more readily available in Supabase. One advantage that Firebase does have is out of the box support for offline-synchronization.
- **Evidence**: The Supabase CLI allows any developer to run backend services locally with one command (`npx supabase start`). This is a huge asset for an open-source project. Supabase is naturally relational with PostgreSQL, and Firebase is natively NoSQL. Firebase does support SQL using SQL Connect, which requires some additional configuration. Offline synchronization is possible with Supabase, but requires more manual configuration than Firebase.
- **What would change our mind**: If offline synchronization became too complex with Supabase, switching to Firebase would become a real consideration. Or, if the scale of the project grows enough and pricing between Supabase and Firebase favor Firebase, that would also merit considering a switch.

7. Risks and open questions

Risk or question	Why it matters	Owner	How we'll find out, and by when
How precisely to represent workout and exercise templates vs workout and exercise sessions/completions in the data model?	Doing so inefficiently can erode DB performance, and add unnecessary overhead to the Data Access component	James	Do research during M1, determine the ideal data model
How to provision Supabase Edge functions and DB	This is where all of the data storage and backend computing will live. Without this, the app does not provide value.	James	Read Supabase docs and run through quick starts for DB and Edge functions during M1
What method of authentication to use	Poor credential selection can impede user experience, or introduce security overhead (which is not strictly negative.)	James	Determine top 3 credential options and weigh pros and cons. Make a selection as part of M1
How to distribute to iOS devices without the $99/year developer profile fee?	Purchasing the developer profile would significantly increase the cost of the project. But, iOS is a major mobile OS that cannot be simply neglected, but must be supported	James	Research distribution methods during M1 and M2, select a long-term solution with a fallback plan.
RepDB's Starter license has limited terms	Misunderstanding the limits could lead to legal liability, necessity to purchase their expensive paid license, or pivot to another exercise catalog	James	Determine and document limits of free version of license during M2, before public demo and release
Will workouts be lost if the final save fails?	Some gyms have spotty wifi, so the final save to the DB failing is a very real possibility. Users losing their logged progress due to this would be a major failure in one of the primary intentions of the app	James	During M2, research methods to backup copies of the workout session and recover if the DB write fails

8. Implementation plan

Milestone	What works when it's done	Depends on	Build hours (estimate)
1	The first core flow (template creation) works, without "one-rep" maximum support	—	12
2	Both core flows (template creation and workout sessions) work and support "one-rep" maximum recording and calculations based off that value.	1	8

Sign-off

Reviewer	Role	Date	Approved
[name]	[role]	[date]	□